Breast Cancer Classification & MLOps Tracking

Leveraging Random Forest & MLflow for Precision Diagnosis

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Executive Summary

A modern approach to clinical diagnostic modeling, combining the power of Random Forest with the transparency of the MLflow lifecycle.

State-of-the-Art Accuracy

96.49%

Model Accuracy

Proven Reliability

Through rigorous feature scaling and iterative model tuning, our Random Forest classifier achieved a near-perfect accuracy rate, ensuring safe and reliable decision support in clinical settings.

The Diagnostic Imperative

Early Detection Saves Lives

Breast cancer remains a leading global health concern. The transition from manual data review to AI-assisted classification allows for:

- Reduced human error in cellular analysis
- Instantaneous processing of complex diagnostics
- Scalable screening across hospital networks

Project Development Lifecycle

Research

Dataset Selection & Hypothesis Generation

Preprocessing

Feature Scaling & Data Augmentation

Modeling

Random Forest & MLflow Integration

Deployment

Flask API & Real-time Inference

Data Preprocessing Strategy

Feature Scaling

Utilized **StandardScaler** to normalize multi-dimensional clinical inputs, preventing bias toward features with larger numeric ranges.

Evaluation Split

Implemented an **80/20 train-test split** to ensure the model generalizes well to unseen patient samples, avoiding overfitting.

Why Random Forest?

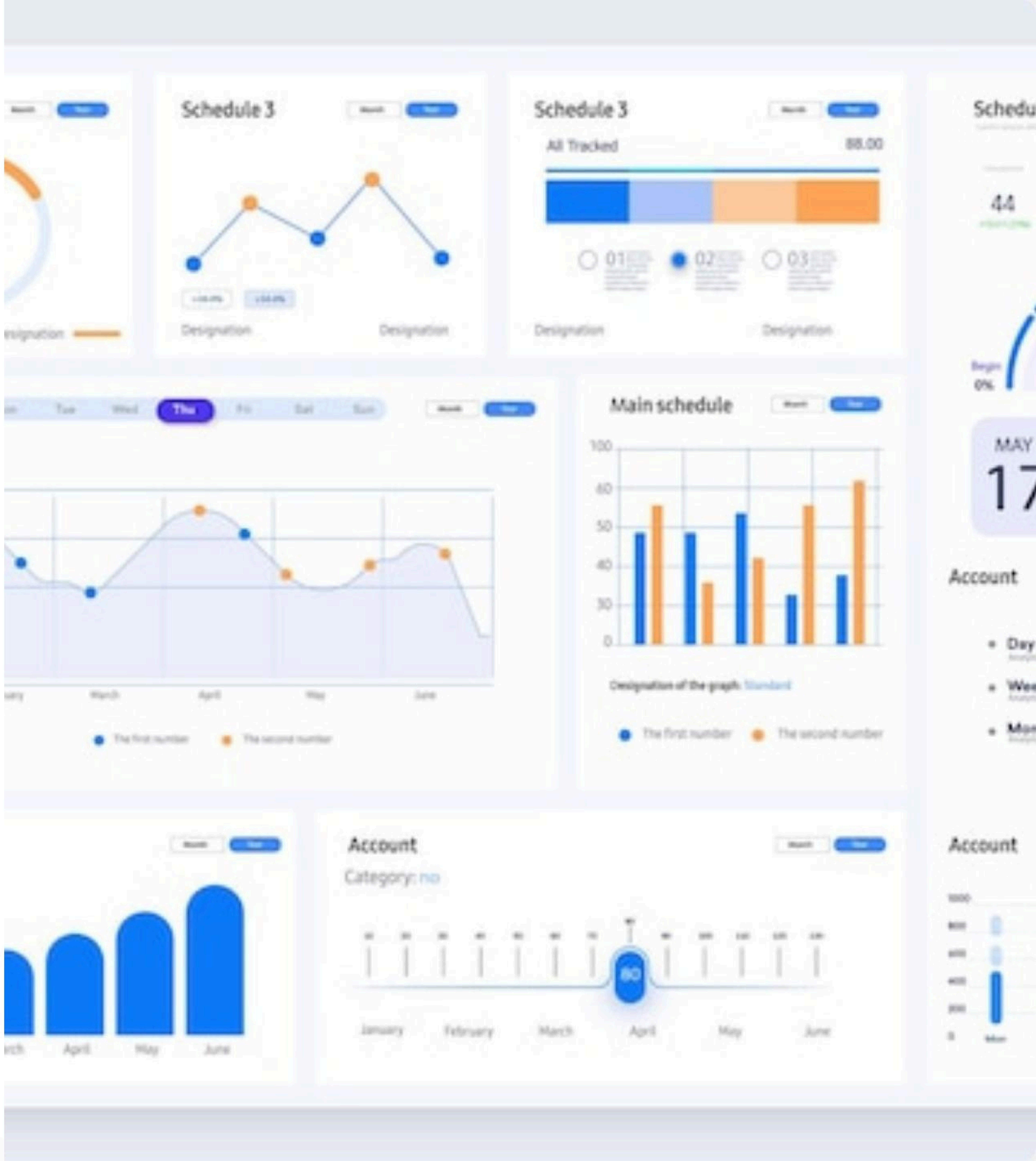
- **Robustness:** Effectively handles non-linear relationships in medical datasets.
- **Feature Importance:** Allows clinicians to see which biological markers contribute most to the diagnosis.
- **Stability:** The ensemble nature of the trees reduces the variance associated with single decision trees.
- **Scalability:** Highly efficient during both training and high-speed prediction phases.

MLOps Transparency

MLflow Integration

Gone are the days of manual spreadsheets. By integrating **MLflow**, we captured every hyperparameter, metric, and artifact automatically.

This ensures 100% reproducibility of the 96.49% accuracy result across any environment.



Performance Summary

Metric / Component	Specification	Result
Classification Model	Random Forest Classifier	Selected for Stability
Accuracy	Final Validation Test	**96.49%**
Tracking Backend	MLflow + SQLite	Persistent History
Artifacts Saved	model.pkl & scaler.pkl	Production Ready

Real-time Deployment Workflow

API Request

Clinician inputs patient symptoms via a secure JSON POST request to our Flask server.

Processing

Data is instantly scaled and passed through the saved Random Forest model.

Prediction

An instant diagnosis (Malignant/Benign) is returned with high-confidence metrics.

The Path Forward: AI-Guard

Applying these same MLOps principles—tracking, scalability, and transparency—to pioneer detection in **Advanced Security & Deepfake Identification**.

> “Early detection isn't just about data; it's about giving families more time. Technology is our greatest tool in that mission.

— *Radwa Yahia, Project Lead*

Questions?

Thank you for your time and attention.

Radwa

King Salman International University (KSIU)

Field Training - Spring 2026

Dr.Saeed Mohsen